

A3-PS005-TC179

Explainable BGP Origin and Subprefix Anomaly Monitor

Team Adam

Partha Chakraborty (Mentor)
Rohan Debnath
Debaditya Das

AIORI Hackathon 3

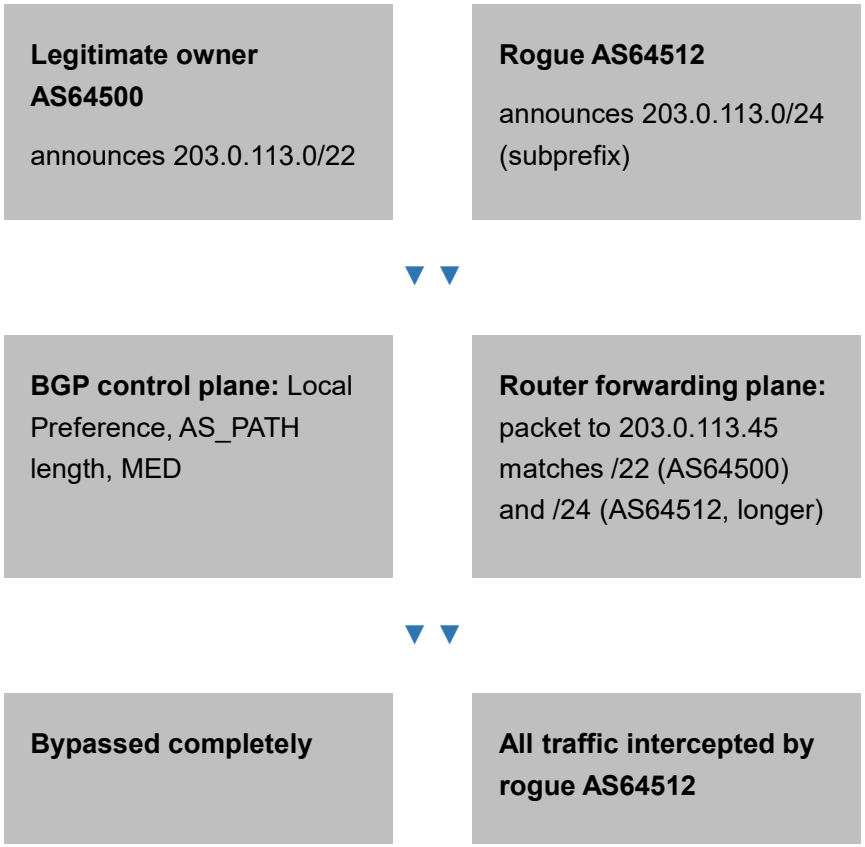
October, 2026

Problem Description

Clear outline of the problem being addressed. Use bullet points for key issues.

- **No BGP authentication:** trust across 70,000+ ASes, no cryptographic proof an AS owns the prefix.
- **Subprefix hijacking:** Longest Prefix Match prefers a /24 over the owner's /22, bypassing Local Preference, AS_PATH length and MED.
- **RPKI blind spot:** over 50–60% of routable prefixes lack ROA coverage (NotFound state).
- **False alarm dilemma:** legitimate MOAS and TE (Anycast DNS, CDN expansion, DDoS scrubbing) cause alert fatigue.
- **Black-box ML:** opaque scores (e.g., 0.93) give NOCs no forensic evidence.

Subprefix LPM Hijack



Background

Foundational concepts necessary to understand the problem and proposed optimization.

- **BGP mechanics:** UPDATEs (NLRI + path attributes) flow Adj-RIB-In → policy filters → Loc-RIB → Adj-RIB-Out.
- **RPKI & RFC 6811 ROV [1]:** Routinator/rpki-client validators output VRPs (Prefix, MaxLength, Authorized ASN); updates are Valid, NotFound, or Invalid (Invalid:AS, Invalid:ML, Invalid:ML-AS).
- **Limitation:** ROV is deterministic, but blind to attacks in NotFound space.
- **Proposed paradigm:** dual-tier retrospective design: Patricia trie baseline, O(k), plus unsupervised Isolation Forest; TreeSHAP yields forensic evidence cards.

RFC 6811 State	Validation Logic	Monitor Handling
Valid	Origin ASN matches AND len <= maxLen	Passed to Tier 2 (ML Monitor)
NotFound	No covering ROA exists in cache	Tier 2 (Isolation Forest + SHAP)
Invalid:AS	Covering ROA exists; Origin ASN !=	Tier 1 Baseline (Instant Alert)
Invalid:ML	Origin matches; prefix len > maxLen	Tier 1 Baseline (Instant Alert)
Invalid:ML-AS	Both Origin ASN and maxLen violated	Tier 1 Baseline (Critical Alert)

Literature & RFC Review

Detailed study of existing proposals, studies, and standards with critical gap analysis.

- **RFC 6811 & RFC 6810 [1], [2]:** standardized ROV, but no defense inside NotFound space.
- **RFC 9319 [3]:** loose maxLength in ROAs enables trivial subprefix hijacking.
- **Supervised ML (Khan et al., 2024 [4]; López et al., 2026 [5]):** SVM/RF/LSTM fail on extreme class imbalance (<0.01% malicious) and zero-day hijacks.
- **Isolation Forests** handle imbalance but lack explainability; **TreeSHAP** gives exact Shapley values in polynomial time (Lundberg et al., 2020 [6]).
- **Tool gap:** BGPStream, RouteViews, RIPE RIS are raw collectors [7]; Cisco Crosswork and ThousandEyes use proprietary heuristics.

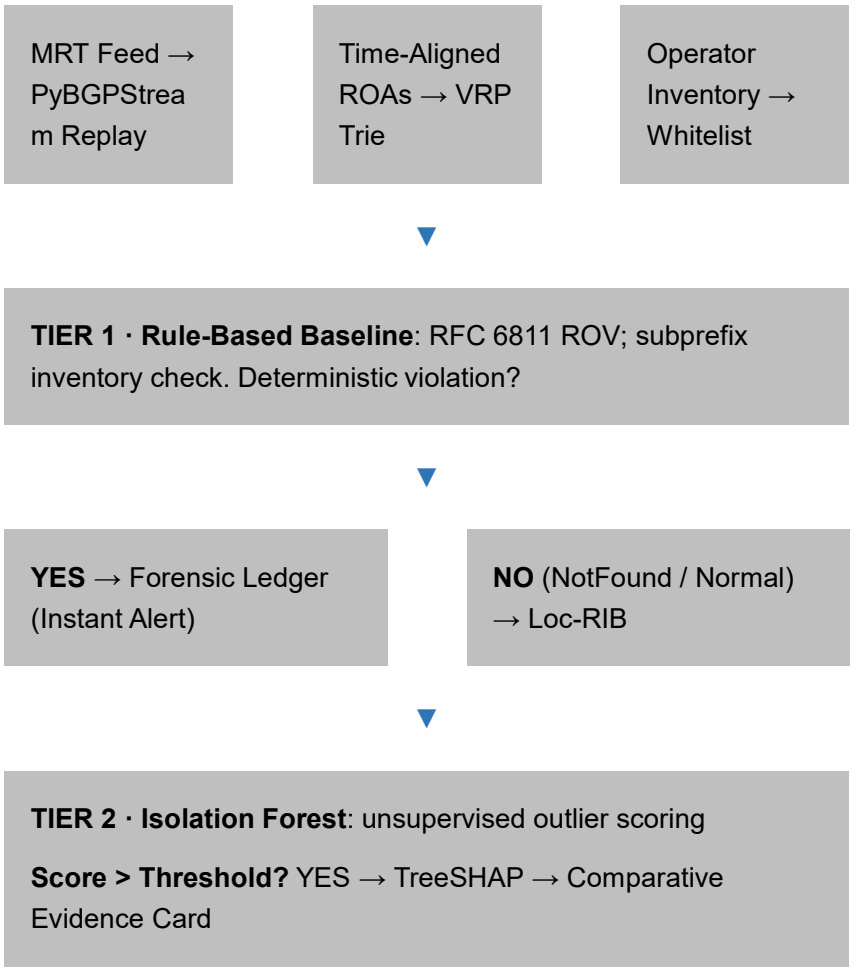
Dimension	RFC 6811 (ROV)	Standard ML (SVM/NN)	Proposed Architecture
RPKI Covered Space	Deterministic (100%)	Probabilistic (~90%)	Deterministic (100%)
NotFound Coverage	Blind (0%)	Detects (~80-85%)	High Recall (~90-92%)
Class Imbalance Handling	N/A (Rule-based)	Poor (High FP rate)	Native (iForest)
Alert Explainability	High (Error code)	Opaque Black-Box	TreeSHAP Attribution
MOAS / Anycast Handling	Brittle (False Pos)	Confused by Spikes	Relational Whitelist
Compute Latency	Sub-millisecond	Variable / High	Gated: <5ms to <50ms

Proposed Solution

Detailed description of the dual-tier explainable detection architecture.

- **Tier 1 (deterministic):** Patricia trie runs RFC 6811 ROV + Relational Operator Inventory; violations alert instantly, bypass Tier 2.
- **Tier 2 (behavioral):** one Isolation Forest on sliding-window features (burstiness, origin persistence, subprefix degree).
- **TreeSHAP:** exact Shapley contributions above θ_anomaly become Evidence Cards.
- **Relational inventory** (primary_asn, secondary_asns, max_deaggregation, policy_state) prevents MOAS false alarms.
- **Monitor-only:** synthetic fault injection for benchmarking; no automated blackholing.

Dual-Tier Monitoring Architecture



AIORI 3 SOLUTION OPTIMIZATION

Optimization Proposed

Optimizations and enhancements, before vs. after.

- **Deterministic gating:** RFC 6811 fast path alerts Invalid:AS/ML/ML-AS in <5 ms, bypassing ML.
- **Relational prefix inventory:** primary and secondary ASNs, TE limits, and states (e.g., DDOS_MITIGATION).
- **In-memory Patricia trie** (O(k) lookup) with temporal sync: no disk I/O; ROA snapshots reload at boundaries.
- **TreeSHAP attribution:** exact feature attributions for Isolation Forest alerts.

Capability / Metric	Before (Conventional)	After (Proposed)	Operational Gain
RPKI NotFound Coverage	Blind to NotFound	Isolation Forest behavioral scoring	Recall ~45% to 88-92%
MOAS & CDN Anycast	Rigid single-origin checks	Relational inventory + temporal tracking	FPR below 4%
Alert Explainability	Opaque score (0.91)	TreeSHAP attribution + containment map	Immediate root cause
Detection Latency	Disk I/O stalls (>150ms)	In-memory Patricia trie fast path	<5ms rules; <50ms ML
Historical Sync	Static/current ROA cache	Time-indexed snapshots synced to MRT clock	100% historical accuracy

AIORI 3 Architecture Design (UML Diagram)

Solution Architecture & Design

DATA INGESTION & STORAGE

- **Historical BGP Archive (MRT)** — RouteViews / RIPE RIS dumps
- **Historical RPKI ROA Snapshots** — VRP cache / trust anchors
- **Operator Prefix Inventory** — relational schema & state

▼

REPLAY & SYNTHETIC INJECTION

- **BGP Replay Engine** — PyBGP deserializer, Unix TS clock
- **Time-Aligned ROA Sync Manager** — dynamic snapshot loader
- **Synthetic Fault Injector** — anomaly injector (origin / subprefix / MaxLength), legitimate injector (Anycast MOAS / TE splits), hidden ground-truth tags

DUAL-TIER DETECTION ENGINE ▼ Tagged BGP UPDATE stream

TIER 1 · Deterministic fast path

- **Patricia Trie ROV Engine** — RFC 6811 origin check, MaxLength check
- **Relational Inventory Validator** — allowed ASNs, deaggregation gate

TIER 2 · Unsupervised behavioral ML (fed NotFound / Valid)

- **Stateful Feature Extractor** — 5-min tumbling window, 8D vector
- → **Isolation Forest Detector** — iTrees scoring, threshold s ≥ 0.80
- → **TreeSHAP Attribution** — exact local explanations

▼ Violation detected

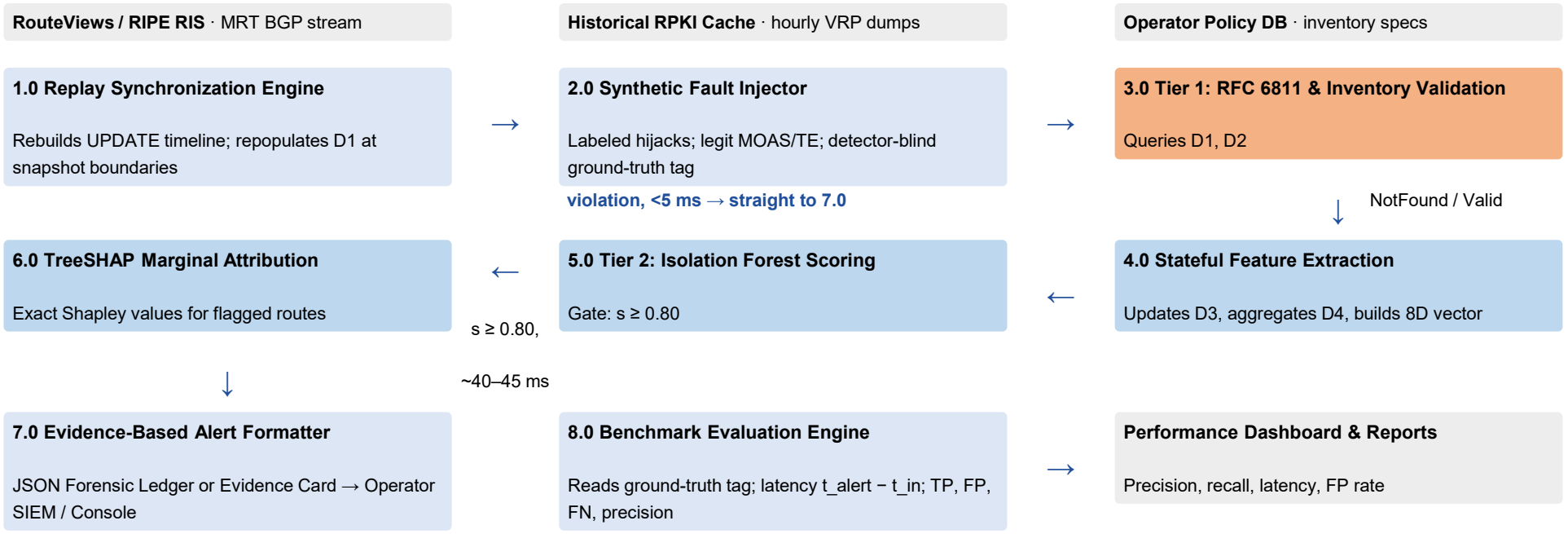
ALERT & BENCHMARKING

- **Evidence-Based Alert Engine** — forensic ledger alerts (Tier 1), comparative evidence cards (Tier 2), JSON-schema payloads
- → **Benchmark Evaluation Engine** — ground-truth tag extraction, precision / recall / F1, latency profiler & FP evaluator

Figure: System Architecture UML Diagram

AIORI 3 Data Flow Diagram

Data Flow Diagram



DATA STORES

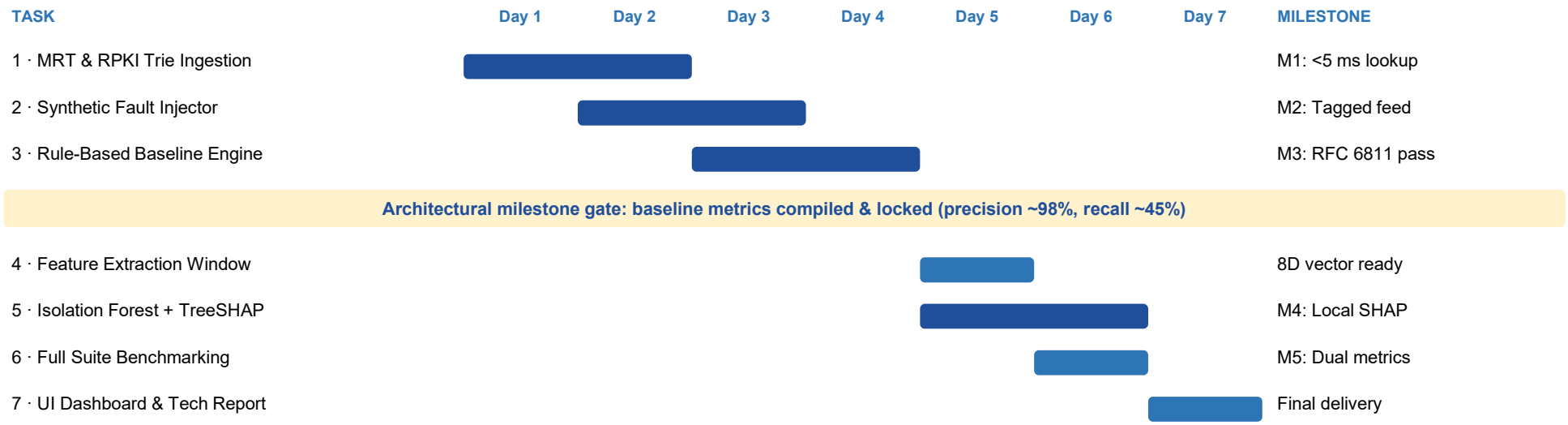
D1 In-Memory Patricia Trie	D2 Relational Operator Inventory	D3 Simulated Loc-RIB State	D4 Sliding Window Buffer	D5 Incident & Evaluation Ledger
read · periodic rebuild	read-only	read · incremental write	5-min tumbling, in-memory stream	append-only JSON log

Figure: Data Flow Diagram (DFD)

AIORI 3 Deployment Timeline

Deployment Timeline

Clearly define the deployment milestones and methodology.



Milestone	Target	Acceptance criteria
M1	End of Day 2	Patricia trie returns correct RFC 6811 states, latency < 5 ms
M2	End of Day 3	Injector produces 5 synthetic classes with hidden tags intact across 100,000 updates
M3 (Gate)	End of Day 4	Rule engine ~98% precision on RPKI-covered space; baseline metrics report locked
M4	End of Day 6	Isolation Forest flags unverified subprefix attacks; TreeSHAP outputs signed feature weights
M5	End of Day 7	Evaluator outputs full confusion matrix; repo ownership transferred to aiori-hackathon

Bibliography

Standards

[1] P. Mohapatra, J. Scudder, D. Ward, R. Bush, R. Austein, “BGP Prefix Origin Validation,” RFC 6811, IETF, Jan. 2013 – *Valid/NotFound/Invalid logic in Tier 1.*

[2] R. Bush, R. Austein, “The Resource Public Key Infrastructure (RPKI) to Router Protocol,” RFC 6810, IETF, Jan. 2013 – *VRP retrieval from RP caches.*

[3] Y. Gilad, S. Goldberg, K. Sriram, J. Snijders, B. Maddison, “The Use of maxLength in the RPKI,” RFC 9319 / BCP 185, IETF, Oct. 2022 – *loose maxLength risks; basis for subprefix detectors.*

[4] Y. Rekhter, T. Li, S. Hares, “A Border Gateway Protocol 4 (BGP-4),” RFC 4271, IETF, Jan. 2006 – *UPDATE format, NLRI, LPM.*

Academic

[5] F. T. Liu, K. M. Ting, Z.-H. Zhou, “Isolation Forest,” Proc. 8th IEEE ICDM, Pisa, 2008, pp. 413–422 – *Tier 2 algorithm.*

[6] S. M. Lundberg, G. G. Erion, H. Chen, et al., “From local explanations to global understanding with explainable AI for trees,” Nature Machine Intelligence, vol. 2, no. 1, pp. 56–67, 2020 – *TreeSHAP.*

[7] D. Gouda et al., “ru-RPKI-ready: The Road Left to Full ROA Adoption,” Proc. ACM IMC, 2025 – *ROA distribution, MOAS behavior.*

[8] A. López et al., “BGP Anomaly Detection: From Feature Engineering to Synthetic Datasets,” UAM Technical Report / IEEE NOMS, 2026 – *time-windowed features, synthetic injection.*

Theoretical grounding

Tier / layer	Reference	Role in our system
Tier 1 Deterministic Fast-Path	RFC 6811 & RFC 9319	Patricia trie matching bit-length and Origin ASN boundaries
Historical Cryptographic Sync	RFC 6810	Snapshot cache re-indexing on hourly replay clock boundaries
Tier 2 Unsupervised Modeling	Liu et al., IEEE ICDM 2008	Path-length isolation scoring over 8D behavioral vectors
Explainable Forensic Evidence	Lundberg et al., Nat. Mach. Intell. 2020	Additive local SHAP feature weight decomposition

Thank You for Your Attention!